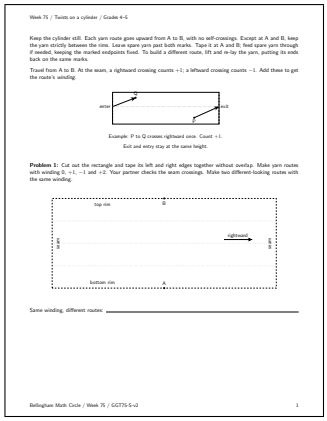
# Week 75 / Twists on a cylinder / Page menu

For adults. Band slices print the pages whose range includes the band, with page and problem numbers unchanged.
K–1: no pages · Grades 2–3: no pages · Grades 4–5: whole packet


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Keep the cylinder still. Each yarn route goes upward from A to B, with no self-crossings. Except at A and B, keep the yarn strictly between the rims. Leave spare yarn past both marks. Tape it at A and B; feed spare yarn through if needed, keeping the marked endpoints fixed. To build a different route, lift and re-lay the yarn, putting its ends back on the same marks.

Travel from A to B. At the seam, a rightward crossing counts +1; a leftward crossing counts −1. Add these to get the route's winding.

Example: P to Q crosses rightward once. Count +1.

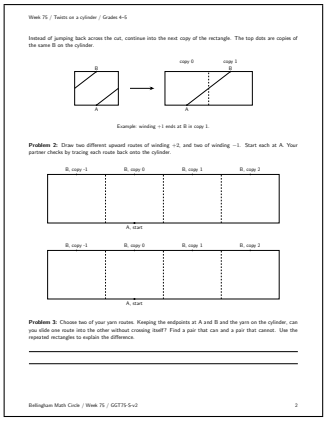
Exit and entry stay at the same height.

**Problem 1:** Cut out the rectangle and tape its left and right edges together without overlap. Make yarn routes with winding 0, +1, −1 and +2. Your partner checks the seam crossings. Make two different-looking routes with the same winding.

Same winding, different routes: _____

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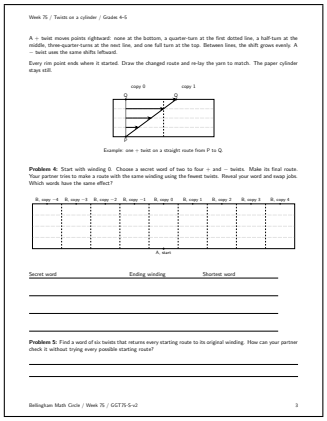
Instead of jumping back across the cut, continue into the next copy of the rectangle. The top dots are copies of the same B on the cylinder.

Example: winding +1 ends at B in copy 1.

**Problem 2:** Draw two different upward routes of winding +2, and two of winding −1. Start each at A. Your partner checks by tracing each route back onto the cylinder.

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A + twist moves points rightward: none at the bottom, a quarter-turn at the first dotted line, a half-turn at the middle, three-quarter-turns at the next line, and one full turn at the top. Between lines, the shift grows evenly. A − twist uses the same shifts leftward.

Every rim point ends where it started. Draw the changed route and re-lay the yarn to match. The paper cylinder stays still.

Example: one + twist on a straight route from P to Q.

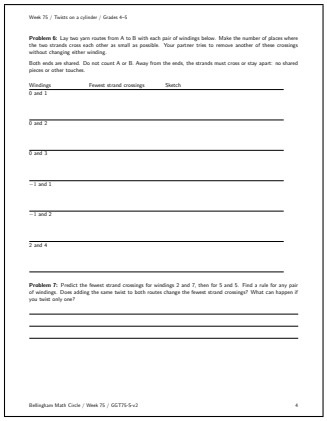
**Problem 4:** Start with winding 0. Choose a secret word of two to four + and − twists. Make its final route. Your partner tries to make a route with the same winding using the fewest twists. Reveal your word and swap jobs. Which words have the same effect?

Secret word Ending winding Shortest word

**Problem 5:** Find a word of six twists that returns every starting route to its original winding. How can your partner check it without trying every possible starting route?

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**Problem 6:** Lay two yarn routes from A to B with each pair of windings below. Make the number of places where the two strands cross each other as small as possible. Your partner tries to remove another of these crossings without changing either winding.

Both ends are shared. Do not count A or B. Away from the ends, the strands must cross or stay apart: no shared pieces or other touches.

| Windings | Fewest strand crossings | Sketch |
|---|---|---|
| 0 and 1 | | |
| 0 and 2 | | |
| 0 and 3 | | |
| −1 and 1 | | |
| −1 and 2 | | |
| 2 and 4 | | |

**Problem 7:** Predict the fewest strand crossings for windings 2 and 7, then for 5 and 5. Find a rule for any pair of windings. Does adding the same twist to both routes change the fewest strand crossings? What can happen if you twist only one?

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